

Second addendum to the 3.0.9/3.1.1 SOLPS-ITER Release Notes

June 12th, 2024

Following various bug reports received from users, some hot fixes, as well as significant code performance improvements, have been applied to the SOLPS-ITER *master* 3.0.9 and *release/3.1.1* code branches. These are:

- Corrected environment files for the MARCONI and SPBSTU clusters (grabbing the right CMake version, in particular).
- Performance improvements for **B2.5** to speed up both serial and OpenMP runs:
 - Only manipulating the parts of the `csigin` array expected to be non-zero.
 - Rewriting the friction force computation loop in `b2sifr_` for better vectorization.
 - Computing the `hy1` cell width function only once when reading the geometry and storing it internally as an array.
 - OpenMP parallelization of loops computing total recycling particle sources per stratum in `b2stbr_phys` and SIMD reductions in the `smin` and `smax` functions.
- Improving the main SOLPS *Makefile* so the proper compiler is always passed to CMake when compiling **Eirene**.
- Replacing the `MPI_PRESENT` main *Makefile* variable with `NO_MPI`, to better handle the situations where the local compiler toolchain does not support MPI.
- Preparing the code for IMAS Data Dictionary changes arriving with DDv4:
 - The convention for closed contours will become to repeat the first point of the contour at the end of the list of points and to drop the `closed` attribute. This affects the **dg2ids** and **ids2dg** programs used to produce *wall* IDSs from **DivGeo** templates and vice-versa.
 - Renaming the `%label` leaves in species description blocks to `%name`.
 - Changing the `_tor` suffix to `_phi` for toroidal IDS quantities (velocities and magnetic field components).
- Bug fixes to allow for the coupled **b2plot** executable to compute wall loadings from a stand-alone **B2.5** result.
- Bug fixes to the *carre* and *triang* scripts to obtain the right grid file number when other(s) are already present and avoid spurious error messages.
- Bug fix for limiter topologies when preparing an IMAS data entry for write-up.
- Compilation fixes when using the `-DASTRA` preprocessor pragma and making the *b2run* script able to handle the **b2mnastra** executable.
- Safeties against too low values of the `DEF_NCUT` variable in *b2mod_dimensions*.
- Other safeties to prevent against usage of uninitialized variables.
- Modifying the **Carre** *Makefile* so the `clean` instruction will also get rid of the *b2mod_dimensions* module file.
- New *setup_baserun_B25_links* script to help in setting up auxiliary files when using **b2plot** to post-process **B2.5** stand-alone run results.
- Clarified some error messages and updated documentation.

The changes above have also been implemented in the 3.0.9 *develop* branch. Use the update scripts as described on the SOLPS-ITER Confluence [page](#) to import them into your own code distribution.